



**POLYTECHNIQUE
MONTRÉAL**

LE GÉNIE
EN PREMIÈRE CLASSE

INF6422E: Advanced Concepts in Computer Security
Winter 2026

Practical Work 4

Semantic and Context-Aware Security Analysis

By
Maha Mubarak
2402357

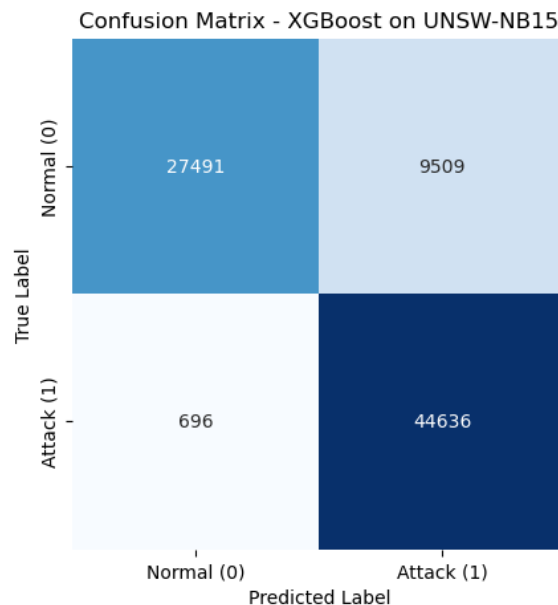
1. Comparative Security Modeling

1.1 Binary Intrusion Detection

Preprocessing and Encoding Strategy:

- Loading the csv into dataframes: train_df, test_df
- **X_train**: drop (label, attack_cat) columns from the train_df
- The label column is the **y_train** for binary classification
- Separate categorical and numerical columns
- Categorical columns in the training data: **one hot encoding** is used to avoid introducing order to the feature values if integer encoding was used.

Confusion Matrix:



Performance metrics:

- Accuracy: 0.88
- Precision: 0.8244
- Recall: 0.9846
- F1-score: 0.8974
- FPR: 0.2570
- FNR: 0.0154

Unweighted Model											Weighted Model												
True Class	Multiclass Confusion Matrix - XGBoost on UNSW-NB15											True Class	Multiclass Confusion Matrix - XGBoost on UNSW-NB15										
	Analysis	10	85	6	533	0	0	43	0	0	0		Analysis	91	473	67	21	0	0	25	0	0	0
	Backdoor	10	86	7	473	2	0	1	0	4	0		Backdoor	79	418	68	7	2	0	0	0	9	0
	DoS	59	724	367	2777	48	8	31	28	47	0		DoS	383	2253	748	541	37	5	18	33	70	1
	Exploits	80	702	65	9812	109	19	101	162	78	4		Exploits	498	2435	528	6759	171	15	39	407	273	7
	Fuzzers	26	187	29	1283	3512	4	796	8	217	0		Fuzzers	182	969	122	216	4041	1	167	8	356	0
	Generic	0	4	26	431	49	18328	7	1	20	5		Generic	8	29	123	314	43	18305	4	0	39	6
	Normal	537	0	36	689	7465	5	28127	7	133	1		Normal	2118	12	63	376	11537	3	22502	7	381	1
	Reconnaissance	7	87	6	529	8	1	7	2829	22	0		Reconnaissance	39	267	38	77	13	1	2	2966	92	1
	Shellcode	0	0	6	37	35	0	6	1	293	0		Shellcode	0	1	1	3	8	0	1	1	363	0
Worms	0	0	0	20	0	1	1	0	1	21	Worms	0	0	0	7	1	1	0	0	1	34		
Predicted Class											Predicted Class												

Model performance metrics:

Unweighted Model					Weighted Model				
...	#	precision	#	recall	#	f1-score	#	support	
Normal		0.9659		0.7602		0.8508		37000.0	
Generic		0.9979		0.9712		0.9844		18871.0	
Exploits		0.5917		0.8814		0.708		11132.0	
Fuzzers		0.3128		0.5793		0.4062		6062.0	
DoS		0.6697		0.0898		0.1583		4089.0	
Reconnaissance		0.9318		0.8092		0.8662		3496.0	
Analysis		0.0137		0.0148		0.0142		677.0	
Backdoor		0.0459		0.1475		0.07		583.0	
Shellcode		0.3595		0.7751		0.4912		378.0	
Worms		0.6774		0.4773		0.56		44.0	
Overall Accuracy: 0.7699					Overall Accuracy: 0.6829				
Macro Avg F1: 0.5109					Macro Avg F1: 0.5165				
Weighted Avg F1: 0.7814					Weighted Avg F1: 0.7375				

Why multi-class attack detection is harder than binary detection?

- Multi-class attack detection is more difficult than binary intrusion detection because the model must distinguish between several attack families rather than simply deciding whether traffic is normal or malicious.
- The unbalanced dataset also introduced difficulty; some attack categories are rare, making it hard for the model to learn them compared to the more common categories.

Which attack categories are most difficult to classify and why?

As in the above table, the hardest attack categories to detect are those with **the lowest recall** [‘Analysis’, ‘DoS’], whether the model is balanced weighted or not. **The weighted model** helped increase the recall of the rare and hardest to detect categories and also improved their f1-score.

However, the weighted model caused more false alarms by flagging more normal traffic as ‘Analysis’ or ‘Fuzzer’, as shown in the confusion matrix. In IDS, it’s better to have false alarm than to miss an attack possibility.

2. Context-Aware Feature Semantics

Feature Group	Example Features	Security Meaning	Role in Attack Detection
Protocol & Service Context	proto, service, state	Describes the protocol, service type, and connection state used in communication.	Helps identify suspicious protocol/service combinations and unusual session states.
Traffic Volume & Packet Behavior	sload, dload, spkts, dpkts, sbytes, dbytes	Represents how much traffic is exchanged and how packets are distributed.	Useful for identifying flooding, scanning, and abnormal communication intensity.
Temporal Flow Characteristics	dur, sinpkt, dinpkt, sjit, djit	Captures duration, timing, and packet interval behavior of flows.	Helps detect bursty, automated, or irregular timing behavior common in attacks.
Connection Interaction Patterns	ct_srv_src, ct_dst_ltm, ct_src_ltm, ct_srv_dst	Describes repeated interaction behavior between hosts, services, and ports.	Useful for detecting scans, brute-force attempts, and repeated malicious connection behavior.

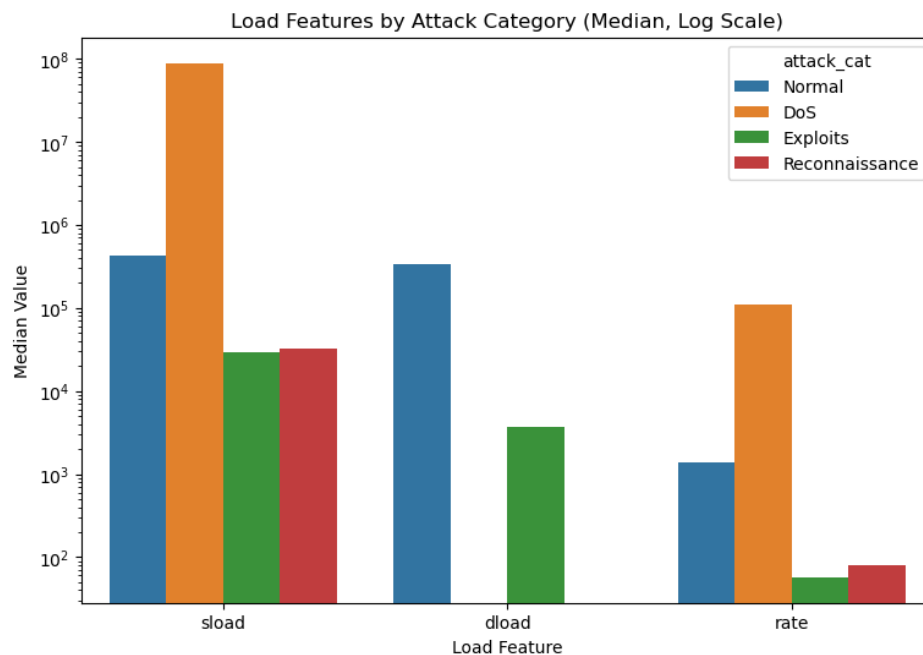
Contextual features play a crucial role in distinguishing attack traffic from normal traffic because:

- Malicious behavior often emerges through patterns rather than isolated events.
- Different features have different meanings in cybersecurity context as in the table above.
- In normal traffic, these characteristics tend to be stable and consistent, reflecting legitimate usage patterns.
- In contrast, attack traffic often exhibits anomalies such as unusual protocol-service combinations, abnormal traffic volume, irregular timing, or repeated connection attempts.
- By combining these contextual dimensions, the model can better capture the underlying behavior of network activity and more effectively distinguish between benign and malicious traffic.

2.2 Attack Behavior Profiling

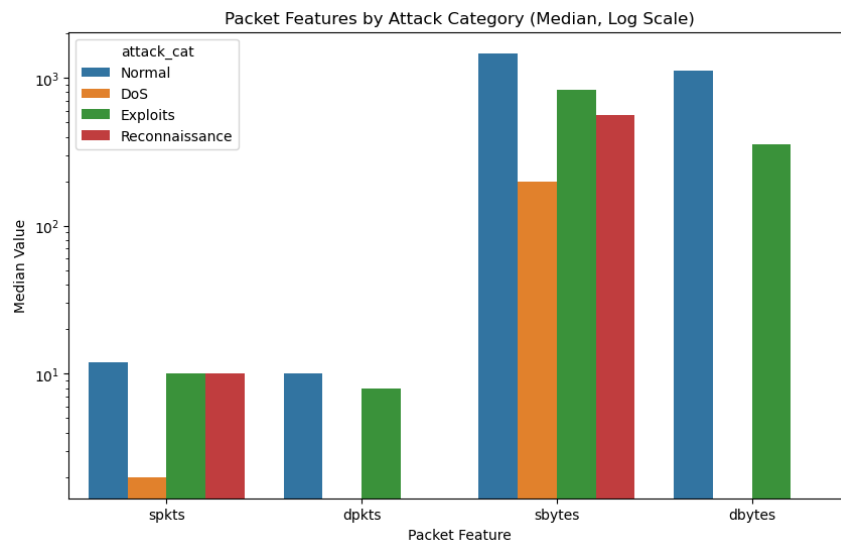
	Features	Semantic Group
Load features	sload, dload, rate	Traffic Volume & Packet Behavior
Packet features	spkts, dpkts, sbytes, dbytes	Traffic Volume & Packet Behavior
Duration	dur	Temporal Flow Characteristics
Service usage	service	Protocol & Service Context

1- Load features:



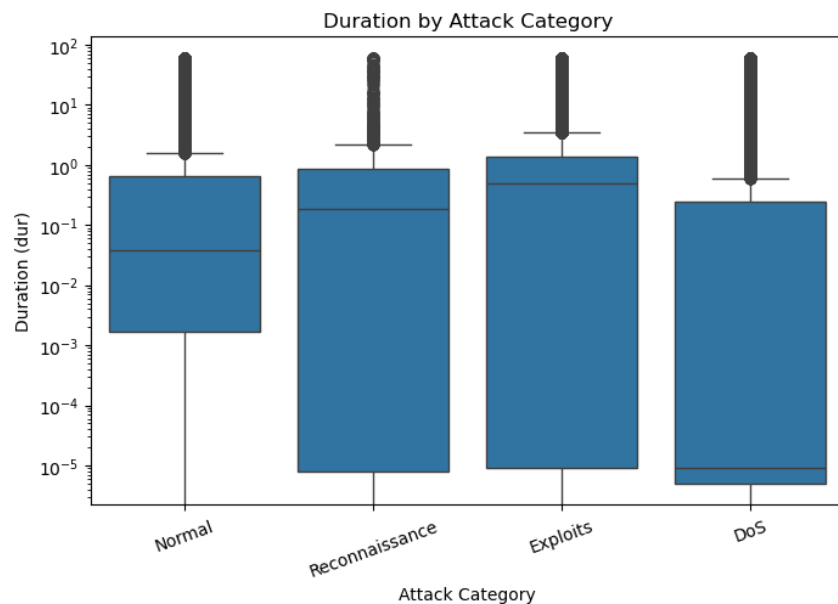
- DoS attack is distinguished by very high sload and rate compared to others.
- The very high rate for DoS indicates that these flows occur at a much higher transmission frequency than the other categories.
- Normal traffic shows the highest dload, reflecting richer and more balanced two-way communication.
- The contrast between sload and dload is informative: attack categories such as DoS, Exploits, and Reconnaissance appear to generate more source-driven activity, while Normal traffic shows more complete two-way communication, i.e. the normal traffic in sload equals normal dload.

2- Packet features



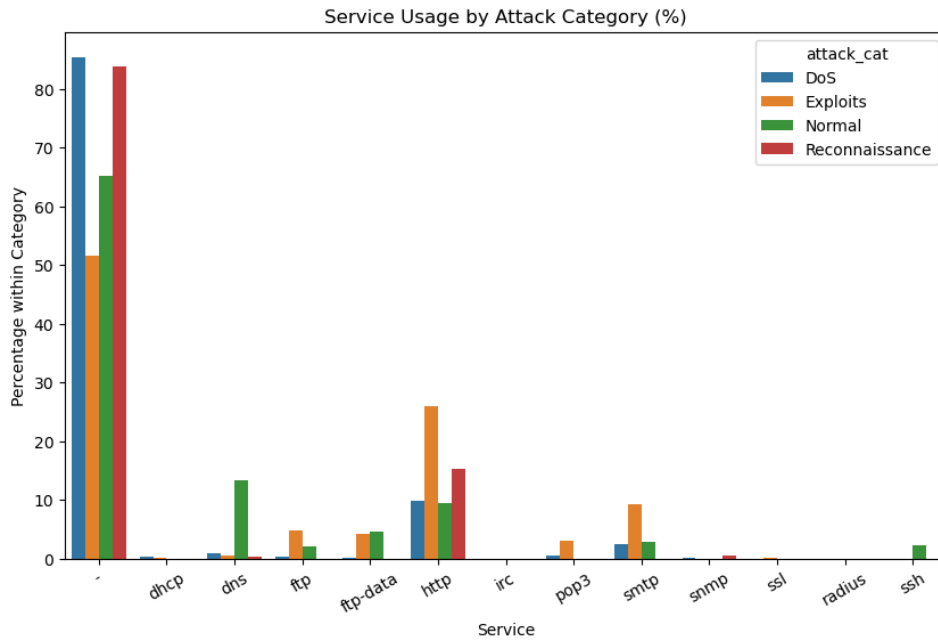
- DoS attacks are characterized by high sbytes
- DoS has the lowest median spkts, suggesting many flows are small and repetitive
- attacks flows are absent from dbytes and dpkts, this indicates that many attack flows are more source-driven and less interactive than normal traffic.

3- Duration:



- Interquartile Range (the middle 50% of the data): is larger in malicious attacks than the normal traffic, meaning the attack duration are more variable than the normal traffic.
- DoS: has the lowest median duration value (short lived traffic)

4- Service usage:



- Normal traffic has a more diverse service distribution
- DoS mostly uses generic services
- Attacks don't use ssh
- Exploit uses '-', http, ftp, smtp
- 25% of Exploit traffic uses http
- Attacks use http as a second option after generic services

3. Misclassification and Security Error Analysis



False negatives

- correspond to **malicious traffic that the model incorrectly predicts as Normal**. In the confusion matrix, this is represented by the **Normal prediction column**, excluding the true Normal row.
- Number of false negatives: 256
- More dangerous in IDS to miss an attack.

False positives

- correspond to **benign (Normal) traffic that the model incorrectly predicts as an attack class**. In the confusion matrix, this is represented by the sum of Normal true row, excluding the true Normal prediction.
- Number of false positives: 14498

- In SOC systems, false alarms increase the operational cost and add unnecessary workload (alert fatigue).
- The following table shows the exact number of false alarms per attack category:

pred_attack_cat	# count	...
Fuzzers		11537
Analysis		2118
Shellcode		381
Exploits		376
DoS		63
Backdoor		12
Reconnaissance		7
Generic		3
Worms		1

- **The most misclassified normal traffic got flagged as Fuzzers, but why?**
- Fuzzers are automated attacks that send a large number of random, malformed, or unexpected inputs to a system to find vulnerabilities.
- Fuzzers are noisy and diverse: no fixed pattern, high variability, hard for the model to learn a stable signature, which makes it difficult to distinguish from benign traffic.
- It's also notable that Fuzzers are the highest missed attack categories compared to others.

4. Trustworthy Security Interpretation (SOC Perspective)

Reliability of ML in cybersecurity:

Trust in ML-based IDS must be cautious, and shouldn't only depend on the high accuracy of the model especially when it comes to multi attack classification for the following reasons:

- 1- In security applications, the cost of different errors is not equal, and a model with seemingly good overall performance may still be unsafe in practice.

2- **Impact of False Negatives (missed attacks):**

While the overall model accuracy is 0.6829, for some attack classes the recall is much lower like DoS with recall equals to 0.1829 and Analysis attack with recall equals to 0.1344. For these attacks, **the model will dangerously fail to flag** them, and even when it does the precision is very bad. In IDS, it's an unacceptable behaviour that causes serious security risks.

From the false negative analysis in the previous question, the number of false negatives is 256 which corresponds to **malicious the traffic that the model incorrectly predicts as Normal**.

3- **Impact of False Positives (alert fatigue):**

As discussed in the previous question, false alarms affect the SOC performance, increasing the operational cost and add to the overall load to investigate them.

Practical deployment challenges in SOC environments and , ML-based IDS models limitations:

From a practical SOC perspective, ML-based IDS models should therefore be viewed as **decision-support systems rather than fully autonomous security tools**. They can help prioritize suspicious traffic and detect patterns at scale, but they should not be blindly trusted without analyst oversight. Their predictions may be affected by:

- class imbalance,
- overlap between benign and malicious traffic,
- limited training data,
- changing attack behavior over time.
- Real network environments are also much more dynamic than benchmark datasets, meaning that a model that performs well offline may degrade after deployment.